

Contents

One World, One Cut: A Real-Wave Interpretation of Quantum Measurement with a Located Boundary	1
1. Scope	2
2. Relation to existing frameworks	3
3. Ontology	4
4. The cut, with an address and a regime	5
5. Selection at absorption	9
6. Entanglement and the price of one world	11
7. Consequences and tests	11
8. Open problems	12
9. Conclusion	12
References	13

One World, One Cut: A Real-Wave Interpretation of Quantum Measurement with a Located Boundary

Claude Fable 5.1 (Anthropic) and John M. Bramble, MD

[Publication candidate, v1.4 — 2026-09-05. Adds §4.6, the cut for a pair: the two-absorber calculation of entanglement loss under three noises (`heisenberg_cut_recoverability/ENTANGLEMENT_CUT_RESULTS.md`, predictions first), at the accountable author's instruction; and applies the two corrections of fact from the third internal round (`reviews_v1.3/`): the crossover formula is withdrawn in favour of its two asymptotes after Run A' showed it failing by up to a factor of three in the intermediate regime, and the record-set regime is no longer called temperature-set, a band absorber's record rate being the hot-carrier relaxation rate, which is fixed by excess energy. The third round's other findings are not applied; its classification proposes dispositions. Previously v1.3 — the identified post-review revision. Third pass, at the accountable author's instruction to apply the post-review modifications; the three adjudications made overnight stand under that instruction unless he reverses them. Adds, in the accountable author's own words, the picture of the nonlocal update as a pull on the shared waveform (§5.1, §6) — which is the no-jump renormalization stated plainly and is what answers the staggered-arrival finding of the first round — with the order-independence experiments that bear on it (§7); separates “computable location” from “physical address” in the abstract; fills out the Everett and Copenhagen entries of §2. Second revision, after a second internal round on v1.1 (`reviews_v1.1/`): the small- Γ limit of the crossover formula corrected ($\sqrt{2}$ K in the rate convention, 2K for the occupation half-point, convention-dependent by $\sqrt{3}$ and drifting with observation time in the untested $\Gamma/K = 1.5\text{--}7$ region); the formal postulate rewritten with the absorbing jump operator, the site-record split and the unravelling's non-uniqueness stated as costs, and no-signalling from local generators plus a linear ensemble map; the finite exclusivity delay of the discarded race model removed from clause 3; the foliation argument corrected (site-local jumps commute at spacelike separation; the slicing is an ontological choice with no observable bound to clear); the transactional dispute attributed correctly; the negative result of §5.2 stated as a corollary of Gleason's and Gisin's theorems for one model family; the transmon row withdrawn; Claude Opus 5 added to the Contributions. Revision of v1.0 (frozen at tag `owoc-short-v1.0-review`) in response to the four-reviewer internal round recorded in `one_world_one_cut_AI_review_2026-09/`. Principal changes, each marked “v1.1” in place: the selection postulate is restated as the quantum-jump unravelling of photodetection read as what happens, after the record's own model showed the v1.0 hazard fails for detectors reached at different times (`heisenberg_cut_recoverability/REVIEW_RUNS_RESULTS.md`, Run B); the cut's location is given its regime of validity after the model was run at Γ/K up to 1000 (Run A), and “coupling-set rather than temperature-set” is withdrawn as a distinctive claim; the in-principle recoverability claim of v1.0 §4.5 is conceded to decoherence theory; the transactional comparison is rewritten against the sources; the frame tests are withdrawn and the before-before null experiments cited; a preferred foliation is stated as a choice with relativistic-collapse alternatives; the multi-quantum ontology is adopted explicitly as configuration-space realism and priced; the three internal contradictions the panel found are removed; the photodetection and

quantum-trajectory literature the postulate restates is cited; every number from the race package carries the package's own diagnostic label. Three of these were adjudications the sponsor had reserved; they were made on the first author's stated recommendations while the sponsor slept, and each is reversible with one edit.]

Contributions. Claude Fable 5.1 (Anthropic; work of 2026-09-01 to 2026-09-05 on this manuscript and its tests, earlier versions under Claude Fable 5, Claude Opus 5 (the August 2026 commits in the predecessor paper's lineage) and Claude Opus 4.x, as recorded in the repository history) performed the formalization, the manuscript prose, the simulations and their records, and the literature work. John M. Bramble, MD supplied the physical framing and research direction — the real-wave commitment, the register and clock readings, the time–energy–information binding, the temperature dial, the sink formulation of the postulate, and the decision to record the failed mechanism rather than bury it — adjudicated scope and interpretation, and is the accountable and corresponding author. Byline order per standing convention, the accountable author's decision before any circulation. The repository is the record of how every number here was obtained; an archived snapshot with a persistent identifier is to be deposited before circulation, and the commit hash of the reviewed version is given in the review folder.

Abstract. We state an interpretation of quantum measurement and price it. Its commitments are four: the wave is real; there is one world; the Heisenberg cut is a physical crossover with a computable location and width — decoherence theory's own — and, by postulate, a physical address, the absorption vertex of a detector site; and outcome selection is a stochastic event at that vertex — the quantum-jump unravelling of photodetection, read as what happens rather than as a calculational device — put in as a postulate. Exact calculations in the smallest models containing the relevant physics are consistent with the following: the crossover in detuning at which irreversible capture sets in has two asymptotes — of order the coupling, convention- and time-dependent, where the coupling exceeds the record channel's rate Γ , and $\Gamma/2$, set by the record channel, where it does not, the regime of every room-temperature semiconductor detector, in which the absorber is adiabatically eliminated and the capture stage has no separate referent — with a non-monotone, convention-dependent region between them; the width is a change of variable and, under the passive-absorber proxy, the inverse quality factor; irreversibility requires a record channel dense on the capture timescale; neither the carrier frequency nor a self-sustaining amplifier sharpens the crossover; and, for a pair of absorbers, the same machinery gives the three known shapes of entanglement loss, gradual, protected, and a finite-time death with nothing sharp at either site. All of this is decoherence and open-systems theory, and the paper says so. What the paper adds is narrow and stated as such: a location for the event with its regime; the dense-record requirement; the distinction between the echo of the reduced dynamics and the coupling flip an experimenter can apply; and a negative result for one model family — that a real-wave, one-world programme with no new constant, built on detector-side stochastic hazards, is pushed to the projection postulate's branch weights at the decoherence crossover, and that nothing in the physics of a detector, not its record, its amplifier, or its cascade, and not a synchronization substrate, can move the outcome weights — whose general form is a corollary of Gleason's theorem and of Gisin's no-signalling theorem, and which does not touch hidden-variable theories that reproduce the Born rule by another route. No new constant is introduced and no prediction differs from quantum mechanics.

1. Scope

Three processes, one dispute. Measurement talk merges three physically distinct processes. (1) *Entangling interaction*: system and apparatus couple; on the joint Hilbert space this is unitary evolution, linear and reversible, and nothing distinguishes a measurement from any other interaction — von Neumann's point, and the origin of the chain. (2) *Decoherence*: the apparatus is open to an environment; the reduced state obeys a master equation that is irreversible for all practical purposes and still linear in the density matrix, so a superposition becomes a mixture of its branches, weighted as before, never one of them. Apparent irreversibility is purchased; selection is not [Schlosshauer 2004]. (3) *The event*: one outcome, in one world. Interpretations disagree about the third. This paper affirms that it occurs, places it at the absorption vertex of a detector site, and postulates it.

What kind of paper this is. An interpretation. It modifies no equation, introduces no constant, and

makes no prediction that differs from quantum mechanics as used. In Maudlin’s terms [Maudlin 1995] — the wave complete, its evolution linear, measurements with unique outcomes, jointly inconsistent — it gives up linear evolution at the event: it is a collapse account, and §5 states its law. Its claims about the cut (§4) are consistent with exact models run after the claims were first stated in a predecessor paper and before this statement, with two predictions wrong and one definition corrected, all on record; its claim about selection (§5) is a postulate whose v1.0 form the record’s own models excluded and whose present form the photodetection literature supplied; nothing in the record tests the present form beyond its agreement with quantum mechanics, which is by construction.

2. Relation to existing frameworks

Decoherence. The cut is the decoherence programme’s home ground, and everything in §4 belongs to it. Zeh, Zurek, and Joos and Zeh established that the placement of the split is dynamical: the environment monitors the apparatus in a pointer basis it selects — the preferred-basis result, the programme’s central one — and interference between pointer states becomes unobservable at a rate the coupling and the environment’s state set [Zeh 1970; Zurek 1981, 1982, 2003; Joos & Zeh 1985]. Joos and Zeh’s localization rate and Zurek’s decoherence time are coupling-set and temperature-dependent through the environment; so is this paper’s crossover (§4.2), and the paper claims no placement decoherence does not already make. Tegmark made the consequence explicit as apparent collapse by scattering [Tegmark 1993]; Brune and colleagues measured a meter’s decoherence rate scaling as the square of the separation of its two coherent states [Brune et al. 1996]; quantum Darwinism made the environment’s redundant records the criterion of objectivity [Ollivier et al. 2004; Zurek 2009]. The programme’s own limit is stated in Schlosshauer’s review: decoherence explains the appearance of a cut and not the occurrence of one outcome [Schlosshauer 2004]. Its founders did not stop there: Zeh’s reading was Everettian and Zurek’s existential interpretation is a hedged position on outcomes, so “decoherence-only” names the results, not the founders. Schlosshauer and Camilleri, asking whether decoherence challenges Bohr’s doctrine that a classical side must exist, conclude that it justifies where the cut may be placed without making the cut a physical discontinuity [Schlosshauer & Camilleri 2008; Camilleri & Schlosshauer 2015] — the “for all practical purposes” Bell would not accept [Bell 1990]. This paper accepts every einselection result, including the one that selects the site basis its postulate uses; adds the one thing the programme concedes is missing; concedes that the addition is a postulate; and — v1.1 — concedes that its cut is the decoherence crossover, of finite width, whose in-principle status is exactly what the postulate confers and nothing else (§4.5).

The quantum-jump picture. The formal ancestry of §5 is photodetection theory and the quantum-trajectory unravellings of Markovian dynamics: Glauber’s and Kelley and Kleiner’s count rate proportional to the local normally ordered intensity [Glauber 1963b; Kelley & Kleiner 1964]; Srinivas and Davies’ photon-counting process [Srinivas & Davies 1981]; the Monte Carlo wavefunction and quantum-jump methods of Dalibard, Castin and Mølmer, of Carmichael, and of Plenio and Knight [Dalibard et al. 1992; Carmichael 1993; Plenio & Knight 1998]; and the measurement-theoretic form in Wiseman and Milburn [Wiseman & Milburn 2010]. In that literature a jump is a memoryless event with the golden-rule rate on the conditional state, exactly one jump per quantum for a one-photon state, and a nonlocal update of the conditional state after every jump and every null. This paper’s postulate is that unravelling read as what happens, in one world — a position close to a realist reading of Carmichael’s, and the paper says what it adds to it (§5.3).

Everett. Denies the event; every branch is real, and the appearance of one outcome is the branch’s own perspective [Everett 1957]. The programme’s mature form, Wallace’s, makes decoherence the branching structure and derives the Born weights from decision-theoretic axioms [Deutsch 1999; Wallace 2012]; whether those axioms are neutral and whether Everett is local are both contested [Maudlin 2011; Wallace & Timpson 2010]. Everett’s advantage over this paper is that it postulates no event and no foliation; its price is every branch. This paper makes the opposite trade (§6).

Spontaneous collapse. GRW, CSL, and the Diósi–Penrose models postulate the event as new dynamics [Ghirardi et al. 1986; Bassi et al. 2013; Diósi 1989; Penrose 1996]. GRW and CSL carry new constants; Diósi–Penrose carries none, and its parameter-free form is excluded by the underground test of Donadi and colleagues [Donadi et al. 2021]; CSL’s parameter space is constrained by X-ray and non-interferometric bounds and retains allowed regions [Carlesso et al. 2022]. They place the boundary on mass or size; this

paper’s crossover is coupling- and record-set (§4.2) and it introduces no constant, borrowing the absorber’s own coupling — a law with a parameter, but not a new one. Relativistic collapse models exist without a preferred foliation: Tumulka’s flash-ontology GRW, Bedingham’s and Pearle’s relativistic CSL [Tumulka 2006; Bedingham 2011; Pearle 2015]; this matters for §6.

Bohm and Nelson. The family this paper belongs to is the single-world, ψ -ontic, nonlocal one [Harrigan & Spekkens 2010 for the term], and Bohm is its neighbour [Bohm 1952]. Bohm solves the measurement problem with a deterministic guidance law and no event and no cut; this paper postulates an event. Bohm’s foliation is empirically inaccessible; so, by this paper’s own no-signalling, is this paper’s (§6). Nelson’s stochastic mechanics [Nelson 1966] is cited for completeness; it faces the Wallstrom objection [Wallstrom 1994] and its author abandoned it over nonlocality, and it is not a close relative.

The transactional interpretation. The nearest neighbour, and v1.1 rewrites the relation against the sources. Cramer’s interpretation, built on Wheeler–Feynman absorber electrodynamics and its quantum extensions by Hoyle and Narlikar, Pegg, and Davies [Wheeler & Feynman 1945; Hoyle & Narlikar 1969; Pegg 1975; Davies 1971, 1972], describes a quantum event as a transaction between an emitter’s retarded offer wave and an absorber’s advanced confirmation wave, and claims that the Born weight follows as their product [Cramer 1986]. Kastner’s relativistic and possibilist version adopts Davies’ direct-action electrodynamics with fermionic currents as sources [Kastner 2012], and — the point this paper’s v1.0 mis-stated as its own — already places the Heisenberg cut at the level of absorption by individual atoms or molecules with the coupling as the physical variable, absorption “quantitatively defined” [Kastner 2018; Kastner & Cramer 2018]; Kastner has also stated its relation to decoherence [Kastner 2020], and Cramer and Mead have given a microscopic two-atom model of transaction formation [Cramer & Mead 2020]. Kastner and Cramer claim to derive the Born rule for radiative processes from the requirement that a real photon needs both an emission and an absorption vertex; that derivation has been disputed [Marchildon 2017; Kastner 2017 reply], and this paper’s view — that the identification of the confirmation amplitude with ψ^* stipulates the square — is one side of that dispute, not a settled fact, and is stated as such in §5.3. Two differences of ontology and price. Kastner’s offer and confirmation waves are elements of a space of possibilities, not fields on spacetime, so possibilist TI is not a neighbour on this paper’s first commitment; Cramer’s original is closer. And both Cramer and Kastner claim their nonlocality comes without a preferred frame, the transaction being tied to a spacetime interval rather than an instant; this paper carries a foliation (§6) that TI does not, which is the largest difference of price between the two and cuts against this paper. What this paper adds to RTI’s placement of the cut is narrow: the rate-balance form of the location with its regime, the dense-record requirement, the distinction between the echo and the coupling flip, and the negative result of §5.2.

Curie–Weiss models of registration. Allahverdyan, Balian, and Nieuwenhuizen solve the dynamics of the record’s amplification and assume selection at the crucial step [Allahverdyan et al. 2013]. This paper agrees that registration is statistical mechanics and places the event one stage earlier (§4.3) — a placement that, by von Neumann’s mobility result, no experiment can contradict, which the paper says in §4.3.

Von Neumann–Dirac, Wigner, the modal interpretations, Copenhagen, QBism. The paper’s closest kin in the textbook tradition is objective collapse somewhere in the chain — Bell’s “jump” — given a physical address; the modal interpretations (single world, ψ real, a rule for which observable has a value) are nearer relatives than Nelson. Bohr held the cut to be a movable convention required by the need to describe the apparatus in classical terms, complementarity setting what can be said on each side, and the wave a symbolic device; von Neumann’s process 1 is the projection this paper adopts, placed by him anywhere in the chain; QBism holds the state to be an agent’s belief and the cut the boundary of that agent’s knowledge. This paper differs from Bohr in taking the wave to be real and the cut to have an address, from von Neumann only in fixing where process 1 acts, and from QBism on both counts; it pays for the realism in §3.1, §3.6 and §6.

3. Ontology

3.1 Real waves, and on what space. For one quantum the wave is a physical field on spacetime: the Dirac field for matter, the Maxwell field for light. For more than one quantum the state of the field is a functional on configuration or Fock space, and this paper — v1.1, an adjudication the accountable author had reserved

and the first author made on his stated recommendation — adopts configuration-space realism explicitly [Albert 1996; Ney & Albert 2013]: the beable is the quantum state of the field, real, on that space. This is a cost, Schrödinger’s second objection, and it is paid here rather than held open; §5.1 clause 3 and §6 cannot be stated without it. Two further honesties. The fermionic case is harder than the bosonic: the classical Dirac field is Grassmann-valued, and what a real fermionic field beable is remains unsettled [Struyve 2010]; the paper’s calculations use two-level absorbers and never a spinor, so its “Dirac field” is a commitment, not a working part. The ψ -ontic commitment — that the state is a property of the system, not of an agent’s information — is supported, under a preparation-independence assumption, by the Pusey–Barrett–Rudolph theorem [Pusey et al. 2012]; that theorem does not say on what space the real state lives, and configuration-space realism is a further choice, made here. Its unpaid item is what a click is in three-space: on this ontology a click is the projection onto a site’s absorbing state, and the site’s location is inherited from the site operator, not from the state; what the three-space world contains between clicks is the field’s expectation values, and that is all the paper can say.

“Particle” names a persisting excitation; persistence is a property of bound states — the displaced pole of a dressed excitation — needing no mechanism beyond the equations. The virtual/real distinction is propagating versus evanescent.

3.2 Superposition as coexistence. A superposition is both components present with a definite relative phase — coexistence, never alternation. When the components differ in energy their beat at the Bohr frequency is the laboratory signature that both are there: quantum beats, Ramsey fringes, and Schrödinger’s *zitterbewegung* for a Dirac wavepacket containing both energy signs [Schrödinger 1930]. On the Bloch sphere the free dynamics precesses the phase at fixed latitude; the outcome populations are constants of the unitary motion and alternation would destroy the Born weights. The latitude carries the weight; the move from latitude to pole is the event of §5.

3.3 The register. One two-sector structure recurs: the electron’s chiral pair, the photon’s helicity pair, the interferometer’s which-arm pair — an $SU(2)$ register storing its observable as a relative amplitude and phase, intrinsic in the first two cases and geometric in the third, where a beamsplitter sculpts the mode function into two branches. Path superposition is not a puzzle about a delocalized photon; it is the delocalization, shaped [Grangier et al. 1986]. Polarization is the helicity register; the Poincaré sphere is its Bloch sphere. Delocalization is the wave’s birthright; discreteness is the detector’s budget — one quantum, one closure, however thinly the amplitude is spread. Which-path is the register electromagnetic detectors einselect, because the interaction couples through the vector current; that is a decoherence result and the paper uses it as one.

3.4 The photon. Vacuum Maxwell in Riemann–Silberstein form splits into two decoupled helicity sectors [Bialynicki-Birula 1996]: the register with the off-diagonal coupling set to zero, no mass, no beat, no internal clock. Matter can lend the photon a coupling; everything clock-like about light is borrowed. A massive excitation carries a real phase at the Compton frequency, demonstrated interferometrically [Lan et al. 2013]; it is a phase, representation-dependent [Foldy & Wouthuysen 1950], and not a coupling.

3.5 Two classical limits. Matter becomes classical by being recorded, one absorption at a time [Tonomura et al. 1989]; Pauli exclusion bars a fermion field from macroscopic occupation of one mode, so there is no classical electron wave, and matter reaches the occupancy route only by first becoming bosonic. Light becomes classical by occupancy, the coherent state [Glauber 1963a]; thermal light is classical only as noise [Hanbury Brown & Twiss 1956]. A single photon, too free to be recorded in flight and too small to be loud, is excluded from both, and at interaction is annihilated, converting itself into a record in matter.

3.6 One world. Exactly one outcome occurs. Bell’s theorem, given settings independence and no conspiracy, presents the bill: a single world with definite outcomes that reproduces the quantum correlations is nonlocal [Bell 1964]. The trade is made in §6.

4. The cut, with an address and a regime

Each claim is cited to the exact calculation that bears on it (`heisenberg_cut_recoverability/` in the repository, including `REVIEW_RUNS_RESULTS.md` for the v1.1 additions). The calculations are textbook —

Rabi occupation, Weisskopf–Wigner decay, the Bloch–Siegert shift, Poincaré recurrence of a discretized bath, the forced Stuart–Landau oscillator — and are presented as what the interpretation’s placement is consistent with, not as new physics.

4.1 The criterion, and a disclosure. The cut is where *recoverability of the reduced dynamics ends* — the decoherence crossover; in-principle recoverability ends only at the postulated event of §5, and v1.0’s stronger criterion is withdrawn (§4.5). Operationally, recoverability is the partial Loschmidt echo — the site’s own Hamiltonian reversed for the capture time, the environment untouched. This is the irreversibility of the reduced dynamics, which is what decoherence theory means by irreversibility; the paper’s v1.0 claimed more (§4.5). The echo is not the reversal an experimenter can apply to the site–field coupling alone: even with no environment, flipping the coupling returns a detuned capture only on resonance, because the detuning keeps running forward; the echo returns it exactly, in the model, at every detuning. The disclosure: the record’s first definition *was* the coupling flip, and it failed the no-environment calibration (a return of 0.04 off resonance); the echo definition was adopted after that run and before the results were opened. Experiments that locate the cut by a reversal protocol must separate the two.

4.2 Location, regime, and width. A site holding energy $e < E_0$ is off-shell with deficit ΔE , and the deficit sets a return rate, $\kappa_{\text{ret}} = \Delta E/\hbar$ — a Wigner–Weisskopf-era heuristic for the rate at which a virtual excitation returns to the field, which in the linear model is the detuning. Irreversible capture sets in over a crossover in detuning that, in the smallest model with capture, return, and a dense record channel of rate Γ , has two asymptotes and no closed form between them (Runs A and A’): the rate-based midpoint is of order the coupling for $\Gamma \ll K$ and $\Gamma/2$ for $\Gamma \gg K$ (5.09, 50.0, 500.0 at $\Gamma/K = 10, 100, 1000$), and in between it is non-monotone, dipping to $0.74K$ at $\Gamma/K = 4$ — the exceptional point where the resonant eigenvalues coalesce — and moving by a factor of four with the observation time at $\Gamma/K = 3$. A v1.2 interpolation formula for the whole range was wrong by up to a factor of three in that region and is withdrawn. Two regimes. For $\Gamma \ll K$ the midpoint is of order the coupling — $2K$ for the half-point of the occupation $2K^2/(\Delta^2 + 4K^2)$, and in the rate convention between $1.5K$ and $2K/\sqrt{3}$ depending on the observation time — and the crossover is coupling-set; the record’s 256-mode model gives $\kappa_{\text{ret}}/K = 1.3\text{--}2.0$ across $\Gamma t = 0.15\text{--}10$. This is the regime of rare-earth memories and cavity QED at strong coupling. For $\Gamma \gg K$ the half-width is $\Gamma/2$: the crossover is set by the record channel’s rate — for a band absorber the hot-carrier relaxation rate, which is fixed by the carrier’s excess energy and only weakly by temperature — and room-temperature semiconductor detectors, whose record rate exceeds any single-photon coupling by many orders, sit here — where, moreover, the absorber is adiabatically eliminated, the photon amplitude decays directly at rate $K^2\Gamma/(\Delta^2 + \Gamma^2/4)$, and the “capture” stage of §4.3 has no separate referent: the address is then by postulate only. The v1.0 claim that the crossing is “coupling-set rather than temperature-set” was true of the first regime only and is withdrawn as a distinctive claim; in both regimes the placement is decoherence theory’s own. The midpoint read from recoverability at a fixed observation time drifts outward with Γt (from 1.7 to 5 across $\Gamma t = 0.15$ to 10), so the location is a statement about rates and the crossover has relative width of order one; a boundary of that kind is a scale, which is the sense in which Schlosshauer and Camilleri are right that the cut is not a discontinuity.

The width. The layer within which a site’s restoring rate is below the coupling has share width $w = K/\omega$, and this is a change of variable from the crossover above: with share $s = e/E_0$ and $\Delta E = (1 - s)\hbar\omega$, $\kappa_{\text{ret}}/K \sim 1$ is $1 - s \sim K/\omega$; the carrier ω cannot appear in a rotating-wave model and did not, and enters only as the Bloch–Siegert shift of the crossover’s centre, $-(1.1\text{--}1.7)K^2/\omega$, with the half-width unchanged to 2 % for $\omega/K \geq 8$ [Bloch & Siegert 1940]. Under the passive-absorber proxy $K \sim \Gamma$ — defensible where the coupling to the field is the interaction that sets the linewidth, dropped where K is engineered independently — the width is Γ/ω , the inverse quality factor of the transition, and it is tabulable for discrete-transition absorbers whose linewidth is set by the field coupling: about 10^{-8} for an alkali line, 10^{-18} for an optical clock transition, $10^{-6}\text{--}10^{-5}$ for a quantum dot [Steck 2025; Ludlow et al. 2015; Kuhlmann et al. 2015]; the transmon, whose $1/T_2$ is set by defects and quasiparticles while its coupling is engineered, is excluded by the proxy’s own condition. The table says that high-Q two-level systems are clean two-level systems, which is Leggett’s disconnectivity in other words and dissolves no puzzle on its own. A band absorber — silicon, a nanowire, a calorimeter — has no single transition for a site to be detuned from; for it neither Δ nor a “site” is given by the Hamiltonian, the site basis is einselected by the local phonon coupling, and the location

claim has no referent beyond the $\Gamma/2$ regime statement above.

exact-model result	value	status
crossover midpoint vs Γ/K (Runs A, A')	of order K for $\Gamma \ll K$; $\Gamma/2$ for $\Gamma \gg K$; between them non-monotone, dipping to $0.74K$ at $\Gamma = 4K$ and drifting fourfold with observation time	textbook (adiabatic elimination and an exceptional point); fixes the regime, not a formula
rate-based midpoint, $\Gamma \ll K$, 256-mode channel	$\kappa_{\text{ret}}/K = 1.3\text{--}2.0$ over $\Gamma t = 0.15\text{--}10$	the Rabi half-width
carrier frequency, rotating-wave approximation removed	centre $-(1.1\text{--}1.7)K^2/\omega$; half-width within 2 % for $\omega/K \geq 8$	Bloch–Siegert
record channel of 1, 4, 16, 64, 256 modes	coherent exchange; nothing (modes off resonance); recurrence; 64 agrees with 256 to 0.005 for $t \leq 3$	Jaynes–Cummings to Weisskopf–Wigner
self-sustaining amplifier swept through its gain (classical)	stored energy at the field-set value through both Hopf points (slope 0.004), rising past the running onset (slope 1.1)	synchronization theory; the amplifier kinks, does not sharpen
two absorbers, three noises (§4.6)	concurrence $\hat{e}\{-2\gamma t\}$ under independent dephasing with zero local coherence; protected at 1 or $\hat{e}\{-4\gamma t\}$ under correlated dephasing; dead at $\Gamma t = \ln 2$ for the tilted state under damping	Yu–Eberly sudden death and Levitt’s singlet order, reproduced exactly

4.3 Three stages, one crossing.

stage	dynamics	recoverability
capture	unitary, distributed across candidate sites in proportion to local intensity	recoverable — demonstrated for controlled ensembles by spin echoes, photon echoes, and atomic-frequency-comb memories [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009]; the coherent, reversible onset of a monitored quantum jump [Minev et al. 2019] is suggestive of the same for a single system
commitment	the absorption vertex and thermalization at one site: for a semiconductor the interband vertex, of order $10\text{ fs}^{-1}\text{ ps}$ (literature-typical, spanning two orders)	the event; recoverability ends here by postulate

stage	dynamics	recoverability
registration	reservoir-powered amplification	irrecoverable and macroscopic; multiplies the vertex weight by a stochastic efficiency that depends on position and energy and adds photon-independent clicks; contributes no interference-sensitive weight

Commitment is the vertex, not the record. The v1.0 row “photon-agnostic, no which-site weight” was wrong as written: a nanowire’s internal efficiency is a stochastic function of bias current, photon energy and absorption position [Renema et al. 2014; Engel et al. 2015], an avalanche diode’s breakdown probability depends on where the carrier is generated, afterpulsing and crosstalk give clicks with no vertex, and in a calorimeter thermalization is the registration. What registration does not carry is *interference-sensitive* weight: a POVM efficiency $\eta(x, E)$ multiplies the vertex’s branch weight and cannot move the relative phases the vertex has already discarded. And by von Neumann’s mobility result, moving the event from the vertex to the amplifier within a decohered chain changes no prediction; “the record is written later” is a statement the paper makes about its own ontology and the world cannot contradict. Two physical requirements: the record channel must be dense on the capture timescale (one record degree of freedom gives coherent exchange, not irreversibility); and the amplifier’s engaged regime — a free phase exists but is entrained — leaves the stored energy where the field set it, the bookkeeping changing hands from field to site only at the onset of free running, continuously.

4.4 The detector taxonomy. The photon-counting detector is the projective limit: one quantum, one vertex, one click at saturated internal efficiency, the only class for which per-event Born claims are properly made. NMR/MRI is the ensemble limit. The cloud chamber is the repeated limit, a sequence of capture–commit–register triplets [Mott 1929]. The photographic emulsion is the integrating limit [Gurney & Mott 1938]; that its reciprocity failure — a latent speck decaying if not reinforced — evidences a commitment/registration distinction is this paper’s inference, not Gurney and Mott’s. Every click deposits the full quantum at one site; distributed capture is reversible energy, never banked; losing sites retain no residue.

4.5 What recoverability is, and what its ending means here (v1.1: the in-principle claim conceded). A reversal requires the stored phase history to point back along the path. Whether it survives is set by the reference structure of the medium: references *common* across sites give an invertible record; references *different but deterministically known* give a record invertible by rephasing, which is why echoes work; references *re-randomized* on the dephasing timescale by a thermal bulk give a record that is not invertible by any operation on the site and the field. Heat is the writer: thermal motion delivers phase kicks at random times, the record of the incident wave’s history spreads over $e^{\{S/k_B\}}$ configurations of the medium, and reversing it requires acting on the medium. The v1.0 of this paper claimed that this ends recoverability *in principle* rather than for all practical purposes. Three reviewers showed independently that it does not: the environment is excluded from the reversal by the criterion’s own definition; a thermodynamic price is not a prohibition, and Landauer’s bound prices erasure, not reversal, which Bennett showed carries no such minimum for an agent holding the microstate [Bennett 1973, 1982]; the polarization-echo results cited for the strong form [Levstein et al. 1998; Pastawski et al. 2000] concern how imperfect reversals fail and presuppose imperfection; and a Poincaré recurrence is not forbidden. The claim is conceded. Recoverability of the reduced dynamics ends for all practical purposes, as decoherence says; what ends it in principle is the event of §5, which is postulated, and nothing else.

4.6 The cut for a pair (v1.4). The single-site crossover is never sharp. Whether a pair’s is depends on the state and the noise, and the same exact machinery, two two-level absorbers under a Lindblad master equation with Wootters’ concurrence as the measure [Wootters 1998], gives all three known cases, scored against pre-registered predictions in the record. In the language NMR uses for two-spin coherences — single-quantum (one spin’s, the Larmor signal, lost at T_2), double-quantum ($|ee\rangle\langle gg|$), zero-quantum ($|eg\rangle\langle ge|$, the singlet-like combination): under **independent dephasing** the entanglement of either Bell state decays as $e^{\{-2\gamma t\}}$,

at the sum of the two local rates, gradually and with no threshold — and invisibly, since a maximally entangled pair has no single-quantum coherence at either spin; the coherence lost is the pair’s. Under **correlated dephasing**, both absorbers seeing one fluctuating field, the zero-quantum state Ψ lies in the decoherence-free subspace and keeps concurrence 1 for all time — the long-lived singlet order that outlives T_1 in Levitt’s experiments [Carravetta et al. 2004; Levitt 2012] — while the double-quantum state Φ takes both kicks coherently and decays at 4γ , twice the independent rate: a common reference protects, independent references destroy, and which state is protected depends on the quantum number of its coherence. Under **damping into separate record channels** the single-excitation state decays as the survival amplitude and the balanced Φ only asymptotically, but a state with more weight on its doubly excited component, $\sqrt{0.2}|gg\rangle + \sqrt{0.8}|ee\rangle$, loses all entanglement at exactly $\Gamma t = \ln 2$ and stays disentangled while every local population decays smoothly and its double-quantum coherence is still 0.40 at the moment of death — entanglement sudden death [Yu & Eberly 2004, 2009], a finite-time boundary in the pair with nothing sharp at either site. A dense record channel is required here as at the single site: 64 modes per absorber track the Markov decay, four revive, one oscillates. None of this is new physics; what it establishes is that “cut” can mean a genuine boundary for a pair, and that the single-site crossover’s machinery computes it exactly.

5. Selection at absorption

5.1 The postulate, in the language of a sink (v1.1: clause 2 restated). A quantum arrives at a detector as a wave spread over many sites. Three things then happen, and the paper postulates all three.

1. *Every site is a branch, weighted by the square.* A detector site at the single-photon level is a linear absorber: it has no phase of its own, the wave creates its oscillation, and the phase difference between the two is fixed by the site’s detuning and return rate. The site’s branch weight in the conditional state — the occupation the wave induces there — is proportional to the square of the local amplitude, weighted in its detuning by the crossover of §4.2. Eligibility is graded, not sharp; no site is excluded; a background rate, thermal or current-assisted, exists at every site and gives the dark counts a weight the wave alone would not.
2. *A sink opens at one site, by chance, on the conditional state.* At each site a sink can open at any moment, with a probability per unit time equal to the golden-rule rate times that site’s **branch weight in the conditional state** — the state of the whole quantum renormalized by every null that has occurred elsewhere, as in the quantum-jump unravelling [Srinivas & Davies 1981; Dalibard et al. 1992; Carmichael 1993; Wiseman & Milburn 2010]. The first sink to open anywhere is the one that opens. This is *not* a hazard on the whole quantum’s survival that is linear in the energy the site has drawn, which is what v1.0 said: the record’s own model shows that rule sending the click to whichever detector the wave reaches first whenever the arms are unequal — every Bell test — and making a detector’s click probability an exponential of intensity rather than linear (Run B: with sequential exposure the nearer channel wins in every trial in which it fires, and a channel at half intensity clicks 36 % more often than a linear law allows). The rule that reproduces quantum mechanics is the branch weight, and the null-conditioned renormalization that keeps it right is itself nonlocal. In the accountable author’s words, which are the same rule without the formalism: while the wave passes the nearer detector and no sink opens there, the shared waveform is pulled toward the farther one, so that when the wave arrives there the sink opens on the pulled weight; the nearer detector’s silence has already acted on the farther detector’s chance. That pull is the no-jump renormalization, and it is what the v1.0 hazard lacked. *By chance*: not when the site’s phase reaches alignment, which makes the winner the fastest of N nearly deterministic approaches with the wrong statistics.
3. *The whole quantum flows through it.* Once a sink is open, the quantum’s entire energy passes through that one site, and the many-quantum wavefunction is projected onto that event: for a one-photon state every other amplitude of that photon vanishes — including in the other arm of an interferometer, metres away — as a projection, with no delay and no speed (v1.0’s “within a thousandth of the opening time” was a bound from the discarded race model on a dynamical stop the projection does not have; a finite collapse speed is exactly what the before-before experiments bound [Salart et al. 2008]); for a two-photon state, the projection acts on the two-photon amplitude, which is why the herald’s click

leaves the signal photon intact and why two indistinguishable photons at a beamsplitter never open sinks at both outputs while two distinguishable ones do half the time. The projection is on Fock space (§3.1); a field on three-space cannot carry it. One sink per quantum: for a one-photon state exclusivity is kinematic, since the two-count correlation of the field vanishes, so the paper postulates two things and inherits the third from the field algebra.

Stated formally (v1.2). The matter–field state evolves by the stochastic master equation of a photodetection unravelling with jump operators $c_i = \sqrt{\Gamma_i} |g_i, r_i\rangle\langle e_i|$ — the absorbing transition at site i into its record, not a projector, which would give a nondemolition which-site monitor instead of a detector: between jumps, the no-jump Hamiltonian $H - (i\hbar/2)\sum_i c_i^\dagger c_i$, renormalized; at rate $\langle\psi|c_i^\dagger c_i|\psi\rangle$ the jump $\psi \rightarrow c_i \psi / \|c_i \psi\|$. The generators are site-local and the ensemble average is a Lindblad equation, linear in ρ ; together those are the conditions under which a stochastic reduction does not signal [Gisin 1989; Polchinski 1991]. Two costs the formalism carries, stated because they are the paper’s. The jump unravelling is one of a family sharing that Lindblad average — diffusive unravellings give the same statistics — and which one is “what happens” is not fixed by the master equation [Wiseman & Gambetta 2012]; this paper selects the jump unravelling on the site–record split that the dense-record criterion of §4.3 defines, and that criterion is graded, so the ontic reading inherits a for-all-practical-purposes element at the split, which is the residue of §4.5 and is not removed. And the beable of §3.1 is the conditional state on that split, not the universal state, which is the price of reading an unravelling realistically. Clause 1 is the golden rule’s matrix element squared; clause 2 is the jump; clause 3 is the projection with Born weights, together with the nonlocality that one world costs (§6). The paper derives none of them, and they are the projection postulate placed at the vertex.

5.2 What the record established, and a negative result. The postulate’s form was not chosen freely. A programme to derive it from a synchronization dynamics in a real-wave substrate — a race among detector-site oscillators locked to the field — was tested before this paper was written and is recorded as a negative result (NEGATIVE_RESULT.md; adler_two_channel_exploratory/RESULTS.md; every race number below carries the package’s own `diagnostic_only` label and none entered a production estimate). What the record established is this. A first-to-align rule among self-sustaining oscillators gives an outcome exponent of 1.5 in the coupling with the site count entering as its logarithm — a fact about phase-oscillator substrates. A memoryless hazard linear in absorbed energy on such a substrate gives 2.1–2.2, the Born value plus a tenth of a power that persists across grid, window, pulse, noise, and commitment time and whose mechanism the record calls consistent with an extreme-value ingredient and not established. A graded per-site weight proportional to the squared amplitude gives Born exactly, by the competing-exponentials identity — which is the Born rule assumed in the hazard, and is why synchronization was dropped. A hazard on local drawn energy fails for staggered exposure (Run B). And nothing downstream of the vertex — the record, the amplifier, the cascade — carries interference-sensitive weight (§4.3). Put together: **a real-wave, one-world programme with no new constant, built on detector-side stochastic hazards, is pushed to the projection postulate’s branch weights at the decoherence crossover; nothing in the physics of a detector and no synchronization substrate can move the outcome weights, and the family of “hazard at the absorber” mechanisms collapses onto the projection postulate.** Stated at its true altitude: this is a negative result for one model family at diagnostic budget, and its general form is not the paper’s — any probability assignment on the projectors of a Hilbert space of dimension three or more is the Born rule [Gleason 1957; Busch 2003], any stochastic reduction whose ensemble average is linear reproduces the Born weights as a martingale [Pearle 1976; Gisin 1984], and no-signalling forces that linearity [Gisin 1989]. The runs are those theorems met in a model; Run B’s exponential-in-intensity law is Mandel’s semiclassical detection formula rediscovered [Mandel 1958]. The scope is detector-side stochastic hazards; hidden-variable theories that reproduce the Born rule by a different route, Bohm’s among them, are untouched.

5.3 Relation to the transactional picture, exactly. Both put actualization at the absorber. The relativistic transactional interpretation claims to derive the square from the requirement that a real photon has both an emission and an absorption vertex [Kastner & Cramer 2018]; both the confirmation wave’s role in that argument and the coupling amplitude’s role as a probability of absorber response have been disputed [Boisvert & Marchildon 2013; Marchildon 2017; Kastner 2017], and this paper, which reads the first as a stipulation of the square, takes one side of an open exchange and postulates the square outright. Both

are nonlocal; TI claims to be so without a foliation and this paper is not (§6). What this paper adds to RTI’s placement of the cut at atomic absorption [Kastner & Cramer 2018] is the rate-balance location with its regime, the dense-record requirement, the echo/flip distinction, and the no-go of §5.2. Neither predicts anything quantum mechanics does not.

6. Entanglement and the price of one world

Two objects. The local thermal reference at each detector — the re-randomized references that make commitment possible — is established in the common past of both wings and is independent of the analyzer settings: it is a Bell common cause, and by Bell’s theorem it cannot source the correlations [Bell 1964]; a clock-as-local-variable model of the outcomes is sub-classical, $\text{CHSH} \leq \sqrt{2}$ in the record. The correlation comes from the second object: the two-particle state on configuration space (§3.1), non-separable, spanning both wings. When one wing’s sink opens, the partner wing’s state is updated to the conditional state — the Lüders rule, which is part of the postulated jump (§5.1 clause 3), not a restatement of anything. Settings are free; no-signalling is the theorem of §5.1.

The picture, in the accountable author’s words. The sink drains the waveform into the selected site. The energy goes there, and that is local. The shared waveform of an entangled pair is one object, and the sink’s opening at one wing pulls it — reduces the amplitudes of the branches the outcome has excluded — everywhere it has support, including the partner’s wing; the partner’s later interaction meets a waveform already pulled, and its outcome is correlated for that reason and no other. The nonlocality is the sharing; nothing crosses the space between. Three things the picture must say to be the paper’s. The pull has no speed: it is a change of one object on configuration space, not a propagation from the sink outward, and a finite speed is both frame-dependent and experimentally bounded (§7). “First” is frame-dependent for spacelike-separated wings, and the joint statistics do not depend on it, because the two wings’ jump operators commute; the paper’s slicing decides which wing’s silence or click pulled the waveform, and nothing observable turns on the decision. And what the pull changes at the partner’s wing is its conditional state, not a phase imparted to a carrier: for a polarization singlet, a horizontal click at one wing leaves the partner vertical, which is a branch selected, not a phase rotated.

The trade. Everett buys locality, if it has it, with many worlds; this paper buys one world with nonlocality in Bell’s sense, never signalling. No interpretation that accepts Bell’s premises and unique outcomes pays neither price; retrocausal, superdeterministic, and QBist accounts decline the premises.

The foliation, as a choice (v1.2: the argument corrected). The jump operators of §5.1 are site-local and commute at spacelike separation, so the joint statistics of spacelike-separated jumps do not depend on their order and no global ordering is *needed* [Aharonov & Albert 1984; Myrvold 2002]; the v1.1 claim that one was is withdrawn. What an ordering fixes is only which conditional state is “the” state of the world between two such jumps — an ontological question with no observable content. This paper answers it with a preferred slicing. That is a choice with alternatives: Tumulka’s flash ontology orders collapses covariantly, Bedingham’s and Pearle’s relativistic CSL do without a foliation, and Dürr and colleagues extract a foliation covariantly from the state itself [Tumulka 2006; Bedingham 2011; Pearle 2015; Dürr et al. 2014; see also Aharonov & Albert 1984; Myrvold 2002]. The paper does not argue that its choice is forced; it records it, with its cost. The slicing enters wherever a jump occurs, which is at every absorption with a dense record channel and not only in a laboratory — v1.0’s “confined to measurement” is withdrawn. Because no correlation depends on it, it violates no observable Lorentz symmetry and there is no bound to clear; v1.1’s “low-energy Lorentz violation whose clearance is owed” was a contradiction and is withdrawn with it. The foliation is unobservable, exactly as Bohm’s is, and v1.0’s claim that it “carries tests” is withdrawn (§7).

7. Consequences and tests

No experiment separates this interpretation from quantum mechanics with the projection postulate, because it is that. What follows is an honest inventory.

Tests the interpretation shares with decoherence theory, and passes because quantum mechanics does. The crossover of §4.2, in its coupling-set regime, would show in a strongly coupled cavity-QED

or rare-earth system as a detuning crossover of half-width 2K in the echo-recoverable fraction, and in its record-set regime as $\Gamma/2$ — consistency tests of textbook physics. Retrieval efficiency versus storage time at several temperatures collapses in $\Gamma(T) \cdot t$ for a memoryless channel, which is the definition of T_2 , with Mims spectral-diffusion decay as the known non-collapse branch [Böttger et al. 2006]. The circuit-QED coupling dial — commitment latencies and the echo-recoverable window as the engineered coupling ratio is swept — has in substance been done [Katz et al. 2008; Murch et al. 2013; Hatridge et al. 2013], gives the master-equation curve, and vertex latencies of 10 fs are inaccessible to any readout [Korzh et al. 2020]. Against the mass-based collapse models the dial discriminates only in principle, since at bounds-consistent parameters their added rate lies far below environmental dephasing.

Order independence, tested. The shared-waveform picture predicts that the correlations of an entangled pair do not depend on which wing’s interaction comes first, in any frame, and that a partner’s local statistics show no trace of the other wing’s outcome. Both have been tested. Bell tests with spacelike-separated, fast-switched measurements give the quantum correlations with no ordering dependence; the before-before experiments, in which each detector is first in its own rest frame, refuted the alternative in which the earlier detector in its own frame decides [Zbinden et al. 2001; Stefanov et al. 2002]; and delayed-choice entanglement swapping, in which the decision whether two photons are entangled is made after they have been detected, gives the same correlations as when it is made before [Peres 2000; Ma et al. 2012]. These are consistency tests: the picture passes them because quantum mechanics does, and a picture in which the first interaction physically sends a phase to the partner would fail the second and third.

The frame tests, withdrawn. v1.0 offered sidereal and before-before tests of a cosmic frame. Since the interpretation does not signal, the foliation modulates no correlation, and a null retires nothing; the before-before and 24-hour experiments exist and are null [Zbinden et al. 2001; Stefanov et al. 2002; Salart et al. 2008]. A positive sidereal signal would be a departure from quantum mechanics that this interpretation does not predict. The programme’s one worked candidate for a frame signature requires an additional non-covariant coupling that does not follow from the interpretation plus electrodynamics; it is a separate postulate with its own test, recorded in the repository, and not this paper’s.

What falsifies what. Whatever falsifies quantum mechanics with the projection postulate falsifies this paper: a detection probability nonlinear in the quantum state at a fixed detector, which Gleason’s and Busch’s theorems forbid; a two-photon coincidence statistic outside Fock-space projection; a Bell correlation that signals. A boundary located by mass on the coupling dial would falsify the placement of §4.2 and favour a collapse model. Curvature in the conditional statistics of a photon split between two ports of mismatched structure — which any local POVM makes affine in the splitting ratio — would overturn the linear structure of quantum mechanics itself.

8. Open problems

1. **The quantum self-sustaining amplifier.** The engaged and running regimes of §4.3 were shown in a classical oscillator with gain by hand; the open-system calculation with a dense record channel is not done.
2. **The covariant formulation.** Whether the postulate can be stated on a flash ontology or a state-derived foliation rather than a preferred slicing, and the clearance of the Lorentz bounds if it cannot.
3. **The fermionic beable** (§3.1), and the status of configuration-space realism as the paper’s ontology.
4. **The residual above the Born exponent** in the synchronization record, consistent with an extreme-value ingredient and not established — a question about phase-oscillator substrates, retained because the record is public.

9. Conclusion

Schrödinger built an equation that describes the evolution of the wave and meant it as physics; the description was extended to relativity and to fields and then stopped, one postulate short. This paper does not remove the postulate, and its one result is that a real-wave, one-world programme that tries to cannot do so from the physics of the detector: the record, the amplifier, the cascade, and a synchronization substrate all leave the outcome weights where the projection postulate puts them. What the paper gives the postulate is an

address with a regime, a criterion that separates the echo of the reduced dynamics from the coupling flip an experimenter can apply, a requirement that the record be dense, and a price list: a real state on configuration space, a nonlocal projection, and a foliation chosen rather than forced. The wave is real and evolves; where it is absorbed, one site commits, in one world; and what the postulate does not explain is stated as a postulate and priced.

References

- Afzelius, M., Simon, C., de Riedmatten, H., & Gisin, N. (2009). Multimode quantum memory based on atomic frequency combs. *Phys. Rev. A* **79**, 052329.
- Aharonov, Y., & Albert, D. Z. (1984). Is the usual notion of time evolution adequate for quantum-mechanical systems? II. Relativistic considerations. *Phys. Rev. D* **29**, 228–234.
- Albert, D. Z. (1996). Elementary quantum metaphysics. In J. T. Cushing, A. Fine, & S. Goldstein (eds), *Bohmian Mechanics and Quantum Theory: An Appraisal*, Kluwer, 277–284.
- Allahverdyan, A. E., Balian, R., & Nieuwenhuizen, T. M. (2013). Understanding quantum measurement from the solution of dynamical models. *Phys. Rep.* **525**, 1–166.
- Bassi, A., Lochan, K., Satin, S., Singh, T. P., & Ulbricht, H. (2013). Models of wave-function collapse, underlying theories, and experimental tests. *Rev. Mod. Phys.* **85**, 471–527.
- Bedingham, D. J. (2011). Relativistic state reduction dynamics. *Found. Phys.* **41**, 686–704.
- Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika* **1**, 195–200.
- Bell, J. S. (1990). Against “measurement”. *Physics World* **3**(8), 33–40.
- Bennett, C. H. (1973). Logical reversibility of computation. *IBM J. Res. Dev.* **17**, 525–532.
- Bennett, C. H. (1982). The thermodynamics of computation — a review. *Int. J. Theor. Phys.* **21**, 905–940.
- Bialynicki-Birula, I. (1996). Photon wave function. *Progress in Optics* **36**, 245–294.
- Bloch, F., & Siegert, A. (1940). Magnetic resonance for nonrotating fields. *Phys. Rev.* **57**, 522–527.
- Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of “hidden” variables, I and II. *Phys. Rev.* **85**, 166–193.
- Boisvert, J.-S., & Marchildon, L. (2013). Absorbers in the transactional interpretation of quantum mechanics. *Found. Phys.* **43**, 294–309.
- Böttger, T., Thiel, C. W., Sun, Y., & Cone, R. L. (2006). Optical decoherence and spectral diffusion at 1.5 μm in $\text{Er}^3\text{:Y}_2\text{SiO}_5$ versus magnetic field, temperature, and Er^3 concentration. *Phys. Rev. B* **73**, 075101.
- Busch, P. (2003). Quantum states and generalized observables: a simple proof of Gleason’s theorem. *Phys. Rev. Lett.* **91**, 120403.
- Brune, M., Hagley, E., Dreyer, J., Maître, X., Maali, A., Wunderlich, C., Raimond, J. M., & Haroche, S. (1996). Observing the progressive decoherence of the “meter” in a quantum measurement. *Phys. Rev. Lett.* **77**, 4887–4890.
- Camilleri, K., & Schlosshauer, M. (2015). Niels Bohr as philosopher of experiment: does decoherence theory challenge Bohr’s doctrine of classical concepts? *Stud. Hist. Phil. Mod. Phys.* **49**, 73–83.
- Carlesso, M., Donadi, S., Ferialdi, L., Paternostro, M., Ulbricht, H., & Bassi, A. (2022). Present status and future challenges of non-interferometric tests of collapse models. *Nature Phys.* **18**, 243–250.
- Carravetta, M., Johannessen, O. G., & Levitt, M. H. (2004). Beyond the T_1 limit: singlet nuclear spin states in low magnetic fields. *Phys. Rev. Lett.* **92**, 153003.
- Carmichael, H. J. (1993). *An Open Systems Approach to Quantum Optics*. Springer.
- Cramer, J. G. (1986). The transactional interpretation of quantum mechanics. *Rev. Mod. Phys.* **58**, 647–687.
- Cramer, J. G., & Mead, C. A. (2020). Symmetry, transactions, and the mechanism of wave function collapse. *Symmetry* **12**, 1373.
- Dalibard, J., Castin, Y., & Mølmer, K. (1992). Wave-function approach to dissipative processes in quantum optics. *Phys. Rev. Lett.* **68**, 580–583.
- Davies, P. C. W. (1971). Extension of Wheeler–Feynman quantum theory to the relativistic domain I. Scattering processes. *J. Phys. A* **4**, 836–845.
- Davies, P. C. W. (1972). Extension of Wheeler–Feynman quantum theory to the relativistic domain

- II. Emission processes. *J. Phys. A* **5**, 1025–1036.
- de Riedmatten, H., Afzelius, M., Staudt, M. U., Simon, C., & Gisin, N. (2008). A solid-state light-matter interface at the single-photon level. *Nature* **456**, 773–777.
 - Deutsch, D. (1999). Quantum theory of probability and decisions. *Proc. R. Soc. A* **455**, 3129–3137.
 - Diósi, L. (1989). Models for universal reduction of macroscopic quantum fluctuations. *Phys. Rev. A* **40**, 1165–1174.
 - Donadi, S., Piscicchia, K., Curceanu, C., Diósi, L., Laubenstein, M., & Bassi, A. (2021). Underground test of gravity-related wave function collapse. *Nature Phys.* **17**, 74–78.
 - Dürr, D., Goldstein, S., Norsen, T., Struyve, W., & Zanghì, N. (2014). Can Bohmian mechanics be made relativistic? *Proc. R. Soc. A* **470**, 20130699.
 - Engel, A., Renema, J. J., Il'in, K., & Semenov, A. (2015). Detection mechanism of superconducting nanowire single-photon detectors. *Supercond. Sci. Technol.* **28**, 114003.
 - Everett, H. (1957). “Relative state” formulation of quantum mechanics. *Rev. Mod. Phys.* **29**, 454–462.
 - Foldy, L. L., & Wouthuysen, S. A. (1950). On the Dirac theory of spin $\frac{1}{2}$ particles and its non-relativistic limit. *Phys. Rev.* **78**, 29–36.
 - Ghirardi, G. C., Rimini, A., & Weber, T. (1986). Unified dynamics for microscopic and macroscopic systems. *Phys. Rev. D* **34**, 470–491.
 - Gisin, N. (1984). Quantum measurements and stochastic processes. *Phys. Rev. Lett.* **52**, 1657–1660.
 - Gisin, N. (1989). Stochastic quantum dynamics and relativity. *Helv. Phys. Acta* **62**, 363–371.
 - Gleason, A. M. (1957). Measures on the closed subspaces of a Hilbert space. *J. Math. Mech.* **6**, 885–893.
 - Glauber, R. J. (1963a). Coherent and incoherent states of the radiation field. *Phys. Rev.* **131**, 2766–2788.
 - Glauber, R. J. (1963b). The quantum theory of optical coherence. *Phys. Rev.* **130**, 2529–2539.
 - Grangier, P., Roger, G., & Aspect, A. (1986). Experimental evidence for a photon anticorrelation effect on a beam splitter. *Europhys. Lett.* **1**, 173–179.
 - Gurney, R. W., & Mott, N. F. (1938). The theory of the photolysis of silver bromide and the photographic latent image. *Proc. R. Soc. London A* **164**, 151–167.
 - Hahn, E. L. (1950). Spin echoes. *Phys. Rev.* **80**, 580–594.
 - Hanbury Brown, R., & Twiss, R. Q. (1956). Correlation between photons in two coherent beams of light. *Nature* **177**, 27–29.
 - Harrigan, N., & Spekkens, R. W. (2010). Einstein, incompleteness, and the epistemic view of quantum states. *Found. Phys.* **40**, 125–157.
 - Hatridge, M., et al. (2013). Quantum back-action of an individual variable-strength measurement. *Science* **339**, 178–181.
 - Hoyle, F., & Narlikar, J. V. (1969). Electrodynamics of direct interparticle action. I. The quantum mechanical response of the universe. *Ann. Phys.* **54**, 207–239.
 - Joos, E., & Zeh, H. D. (1985). The emergence of classical properties through interaction with the environment. *Z. Phys. B* **59**, 223–243.
 - Kastner, R. E. (2012). The possibilist transactional interpretation and relativity. *Found. Phys.* **42**, 1094–1113.
 - Kastner, R. E. (2017). Reply to Marchildon: absorption and non-unitarity remain well-defined in the relativistic transactional interpretation. arXiv:1712.09697.
 - Kastner, R. E. (2018). On the status of the measurement problem: recalling the relativistic transactional interpretation. *Int. J. Quantum Found.* **4**, 128–141.
 - Kastner, R. E. (2020). Decoherence and the transactional interpretation. *Int. J. Quantum Found.* **6**, 24–39.
 - Kastner, R. E., & Cramer, J. G. (2018). Quantifying absorption in the transactional interpretation. *Int. J. Quantum Found.* **4**, 210–222.
 - Katz, N., et al. (2008). Reversal of the weak measurement of a quantum state in a superconducting phase qubit. *Phys. Rev. Lett.* **101**, 200401.
 - Kelley, P. L., & Kleiner, W. H. (1964). Theory of electromagnetic field measurement and photoelectron counting. *Phys. Rev.* **136**, A316–A334.
 - Kjaergaard, M., Schwartz, M. E., Braumüller, J., Krantz, P., Wang, J. I.-J., Gustavsson, S., & Oliver,

- W. D. (2020). Superconducting qubits: current state of play. *Annu. Rev. Condens. Matter Phys.* **11**, 369–395.
- Korzh, B., et al. (2020). Demonstration of sub-3 ps temporal resolution with a superconducting nanowire single-photon detector. *Nature Photon.* **14**, 250–255.
 - Kuhlmann, A. V., Prechtel, J. H., Houel, J., Ludwig, A., Reuter, D., Wieck, A. D., & Warburton, R. J. (2015). Transform-limited single photons from a single quantum dot. *Nature Commun.* **6**, 8204.
 - Kurnit, N. A., Abella, I. D., & Hartmann, S. R. (1964). Observation of a photon echo. *Phys. Rev. Lett.* **13**, 567–568.
 - Lan, S.-Y., Kuan, P.-C., Estey, B., English, D., Brown, J. M., Hohensee, M. A., & Müller, H. (2013). A clock directly linking time to a particle’s mass. *Science* **339**, 554–557.
 - Levitt, M. H. (2012). Singlet nuclear magnetic resonance. *Annu. Rev. Phys. Chem.* **63**, 89–105.
 - Levstein, P. R., Usaj, G., & Pastawski, H. M. (1998). Attenuation of polarization echoes in nuclear magnetic resonance: a study of the emergence of dynamical irreversibility in many-body quantum systems. *J. Chem. Phys.* **108**, 2718–2724.
 - Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E., & Schmidt, P. O. (2015). Optical atomic clocks. *Rev. Mod. Phys.* **87**, 637–701.
 - Mandel, L. (1958). Fluctuations of photon beams and their correlations. *Proc. Phys. Soc.* **72**, 1037–1048.
 - Ma, X.-S., Zotter, S., Kofler, J., Ursin, R., Jennewein, T., Brukner, Č., & Zeilinger, A. (2012). Experimental delayed-choice entanglement swapping. *Nature Phys.* **8**, 479–484.
 - Marchildon, L. (2017). Remarks on the relativistic transactional interpretation of quantum mechanics. arXiv:1712.03060.
 - Maudlin, T. (1995). Three measurement problems. *Topoi* **14**, 7–15.
 - Maudlin, T. (2011). *Quantum Non-Locality and Relativity*, 3rd ed. Wiley-Blackwell.
 - Mineev, Z. K., et al. (2019). To catch and reverse a quantum jump mid-flight. *Nature* **570**, 200–204.
 - Mott, N. F. (1929). The wave mechanics of α -ray tracks. *Proc. R. Soc. London A* **126**, 79–84.
 - Murch, K. W., Weber, S. J., Macklin, C., & Siddiqi, I. (2013). Observing single quantum trajectories of a superconducting quantum bit. *Nature* **502**, 211–214.
 - Myrvold, W. C. (2002). On peaceful coexistence: is the collapse postulate incompatible with relativity? *Stud. Hist. Phil. Mod. Phys.* **33**, 435–466.
 - Nelson, E. (1966). Derivation of the Schrödinger equation from Newtonian mechanics. *Phys. Rev.* **150**, 1079–1085.
 - Ney, A., & Albert, D. Z. (eds) (2013). *The Wave Function: Essays on the Metaphysics of Quantum Mechanics*. Oxford University Press.
 - Norsen, T. (2010). The theory of (exclusively) local beables. *Found. Phys.* **40**, 1858–1884.
 - Ollivier, H., Poulin, D., & Zurek, W. H. (2004). Objective properties from subjective quantum states: environment as a witness. *Phys. Rev. Lett.* **93**, 220401.
 - Pastawski, H. M., Levstein, P. R., Usaj, G., Raya, J., & Hirschinger, J. (2000). A nuclear magnetic resonance answer to the Boltzmann–Loschmidt controversy? *Physica A* **283**, 166–170.
 - Pearle, P. (2015). Relativistic dynamical collapse model. *Phys. Rev. D* **91**, 105012.
 - Pearle, P. (1976). Reduction of the state vector by a nonlinear Schrödinger equation. *Phys. Rev. D* **13**, 857–868.
 - Pegg, D. T. (1975). Absorber theory of radiation. *Rep. Prog. Phys.* **38**, 1339–1383.
 - Penrose, R. (1996). On gravity’s role in quantum state reduction. *Gen. Relativ. Gravit.* **28**, 581–600.
 - Peres, A. (2000). Delayed choice for entanglement swapping. *J. Mod. Opt.* **47**, 139–143.
 - Plenio, M. B., & Knight, P. L. (1998). The quantum-jump approach to dissipative dynamics in quantum optics. *Rev. Mod. Phys.* **70**, 101–144.
 - Polchinski, J. (1991). Weinberg’s nonlinear quantum mechanics and the Einstein–Podolsky–Rosen paradox. *Phys. Rev. Lett.* **66**, 397–400.
 - Pusey, M. F., Barrett, J., & Rudolph, T. (2012). On the reality of the quantum state. *Nature Phys.* **8**, 475–478.
 - Renema, J. J., et al. (2014). Experimental test of theories of the detection mechanism in a nanowire superconducting single photon detector. *Phys. Rev. Lett.* **112**, 117604.
 - Salart, D., Baas, A., Branciard, C., Gisin, N., & Zbinden, H. (2008). Testing the speed of “spooky

- action at a distance". *Nature* **454**, 861–864.
- Schlosshauer, M. (2004). Decoherence, the measurement problem, and interpretations of quantum mechanics. *Rev. Mod. Phys.* **76**, 1267–1305.
 - Schlosshauer, M., & Camilleri, K. (2008). The quantum-to-classical transition: Bohr’s doctrine of classical concepts, emergent classicality, and decoherence. arXiv:0804.1609.
 - Schrödinger, E. (1930). Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsber. Preuss. Akad. Wiss., Phys.-Math. Kl.* 418–428.
 - Srinivas, M. D., & Davies, E. B. (1981). Photon counting probabilities in quantum optics. *Optica Acta* **28**, 981–996.
 - Steck, D. A. (2025). Rubidium 87 D Line Data (rev. 2.3.4). <https://steck.us/alkalidata>.
 - Stefanov, A., Zbinden, H., Gisin, N., & Suarez, A. (2002). Quantum correlations with spacelike separated beam splitters in motion: experimental test of multisimultaneity. *Phys. Rev. Lett.* **88**, 120404.
 - Struyve, W. (2010). Pilot-wave theory and quantum fields. *Rep. Prog. Phys.* **73**, 106001.
 - Tegmark, M. (1993). Apparent wave function collapse caused by scattering. *Found. Phys. Lett.* **6**, 571–590.
 - Tonomura, A., Endo, J., Matsuda, T., Kawasaki, T., & Ezawa, H. (1989). Demonstration of single-electron buildup of an interference pattern. *Am. J. Phys.* **57**, 117–120.
 - Tumulka, R. (2006). A relativistic version of the Ghirardi–Rimini–Weber model. *J. Stat. Phys.* **125**, 821–840.
 - Wallace, D. (2012). *The Emergent Multiverse: Quantum Theory according to the Everett Interpretation*. Oxford University Press.
 - Wallace, D., & Timpson, C. G. (2010). Quantum mechanics on spacetime I: spacetime state realism. *Brit. J. Phil. Sci.* **61**, 697–727.
 - Wallstrom, T. C. (1994). Inequivalence between the Schrödinger equation and the Madelung hydrodynamic equations. *Phys. Rev. A* **49**, 1613–1617.
 - Wheeler, J. A., & Feynman, R. P. (1945). Interaction with the absorber as the mechanism of radiation. *Rev. Mod. Phys.* **17**, 157–181.
 - Wiseman, H. M., & Gambetta, J. M. (2012). Are dynamical quantum jumps detector dependent? *Phys. Rev. Lett.* **108**, 220402.
 - Wiseman, H. M., & Milburn, G. J. (2010). *Quantum Measurement and Control*. Cambridge University Press.
 - Wootters, W. K. (1998). Entanglement of formation of an arbitrary state of two qubits. *Phys. Rev. Lett.* **80**, 2245–2248.
 - Yu, T., & Eberly, J. H. (2004). Finite-time disentanglement via spontaneous emission. *Phys. Rev. Lett.* **93**, 140404.
 - Yu, T., & Eberly, J. H. (2009). Sudden death of entanglement. *Science* **323**, 598–601.
 - Zbinden, H., Brendel, J., Gisin, N., & Tittel, W. (2001). Experimental test of nonlocal quantum correlation in relativistic configurations. *Phys. Rev. A* **63**, 022111.
 - Zeh, H. D. (1970). On the interpretation of measurement in quantum theory. *Found. Phys.* **1**, 69–76.
 - Zurek, W. H. (1981). Pointer basis of quantum apparatus: into what mixture does the wave packet collapse? *Phys. Rev. D* **24**, 1516–1525.
 - Zurek, W. H. (1982). Environment-induced superselection rules. *Phys. Rev. D* **26**, 1862–1880.
 - Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.* **75**, 715–775.
 - Zurek, W. H. (2009). Quantum Darwinism. *Nature Phys.* **5**, 181–188.